CSL Lab 6:
Speed control of a DC Motor

General reminders

- Last lab today!
- Please complete a course evaluation – we listen to your feedback!
  - Experience making connection between ModCon and a real-world application
    - without “too” much theory
- Quiz in the last week
  - Duration 45 minutes (most will finish in about 20-30 minutes)
  - Covers major topics, with emphasis on lab-specific knowledge
  - Any make-ups will be oral
  - Pen and Paper exam – no crib sheets, no calculators, no devices out at any time

Review

- Lab 1: Solving ODEs numerically
  - Euler's method
  - Expressing a set of ODEs in a block diagram
- Lab 2: Modeling a DC motor
  - Using 1st order transient response data to identify model parameters
  - Using steady state measurements to identify model parameters
  - Non-linear (non-ideal) behavior in the DC motor system
- Lab 3: Frequency response
  - Relationship between sinusoidal input and output
  - Extracting frequency response from measured data
  - Using bode plots; reading and converting log scale
- Lab 4: Feedback control basics
  - Drawing a block diagram of a feedback system
  - Performance objectives for control
  - Writing the control law for a given compensator
  - Typical reasons for using P, I, and D control
- Lab 5: Motor position control
  - Implementing feedback control in Simulink (function of different blocks)
  - Steady state error due to non-ideal behavior
- Lab 6: Motor speed control
  - PI compensator: U = Kp*E + Ki*int(E)
  - Process to design PI compensator using Root Locus
  - Implementing PI control

Prelab 6: Complete together using SISOTOOL

1. Design Proportional controller for speed control
2. Address Stead State Error with a pure Integrator
3. Improve speed of response with a zero term

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Perform 4 different feedback control experiments:

1.1 Proportional

1.2 Proportional + 0.5 volt offset

2. Proportional with pure integral (+0.5 volt offset)

3. Proportional with pole origin plus a zero somewhere (+0.5 volt offset)

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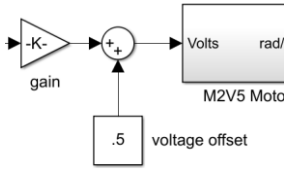
- Implementing your Compensator in Simulink

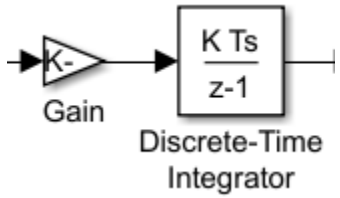
1.1 Proportional: K_1

1.2 Proportional with offset



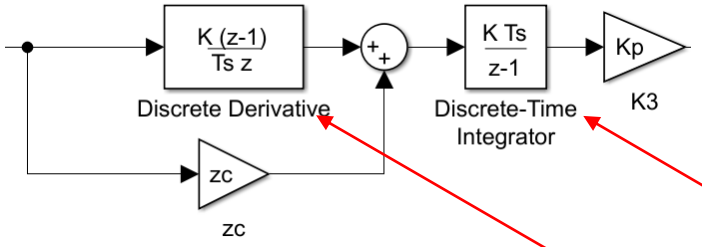
This is just $\frac{1}{s}$
- no need to edit

2. Proportional with pure integral: $\frac{K_2}{s}$



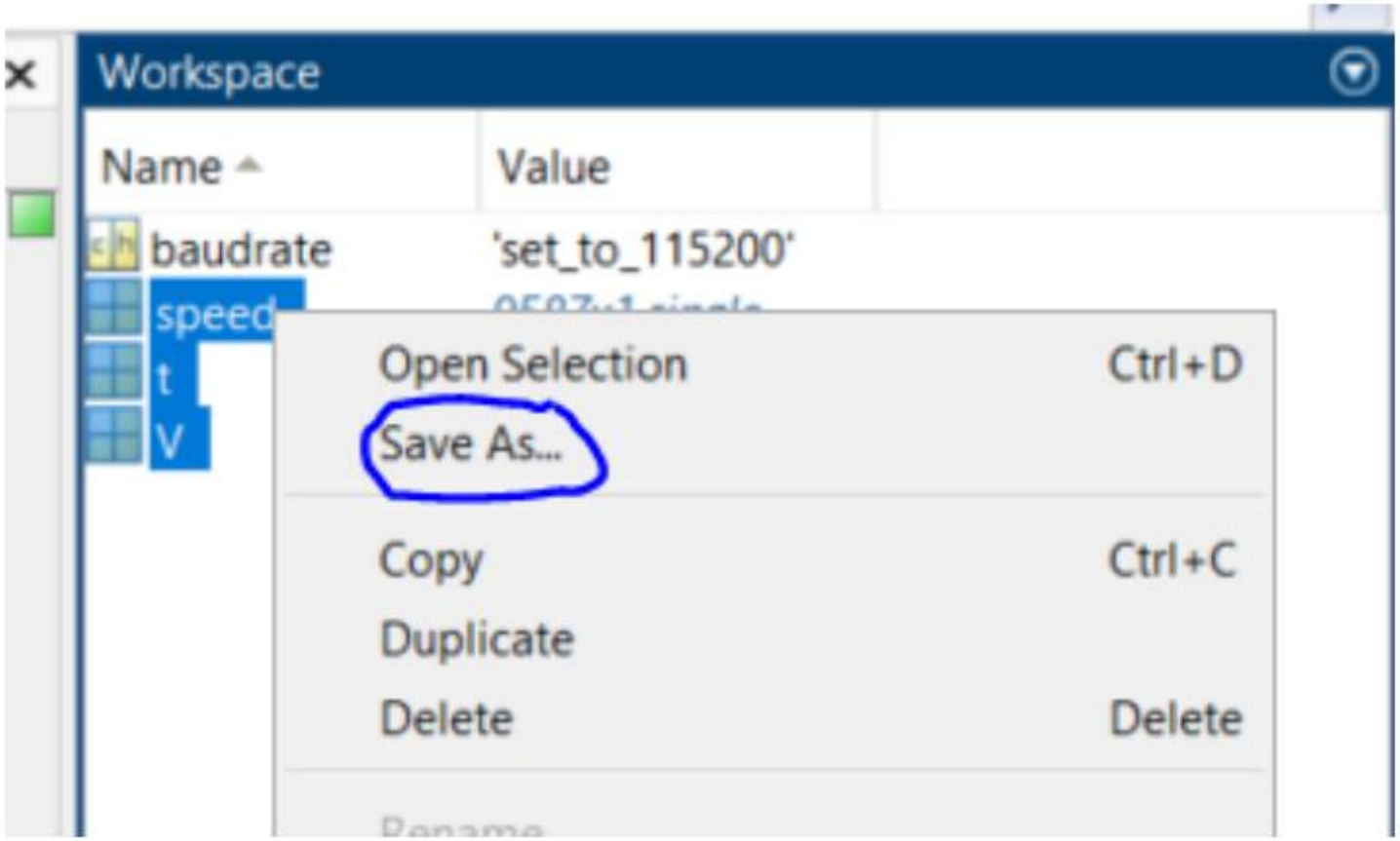
Each compensator is different, so each K is different!

3. Proportional with pole origin plus a zero somewhere: $\frac{K_3(s+z_c)}{s}$



Do NOT edit this block – it is just an integral or "1/s"

Do NOT edit this block – it is just a derivative or "s"


Workspace
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Copy
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Duplicate
Delete
Delete

Don’t forget

- Course Policy: No Webex, discord, email, phones, headphones, social media etc. allowed during labs